

Simulation-Based Assessment of Decarbonization Alternatives in Container Terminals

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Abstract—The lack of open-source platforms capable of integrated operational modeling and multi-scenario decarbonization analysis, often hinders data-driven decision-making in the maritime sector. To address this gap, this paper presents an open-source, multi-agent, discrete-event simulator capable of accurately forecasting the energy consumption associated with the diverse assets and activities within a container terminal. The tool's modular architecture enables transparent evaluations of operational strategies and decarbonization alternatives by allowing users to systematically modify inputs or alter embedded energy modules. The tool's capabilities were validated through a case study of a medium-sized Portuguese container terminal. For this particular port, findings indicate that installing three onshore power supply (OPS) units and fully electrifying the internal truck fleet yields the most substantial emission reductions. However, these interventions result in a two-fold increase in daily electricity demand, potentially straining grid capacity. This finding underscores that the effectiveness of terminal electrification as a decarbonization strategy ultimately depends on a simultaneous transition to a decarbonized and secure energy supply.

Index Terms—container terminal, decarbonization, electrification, multi-agent discrete-event simulation, python

I. INTRODUCTION

In recent years and motivated by the implementation of increasingly stringent environmental regulations, ports have faced significant challenges in reducing carbon emissions and improving energy efficiency [1]. Following the latest International Maritime Organization and the European Commission directives and regulations, port authorities have been committed to the reduction of greenhouse gas (GHG) emissions and to meeting the European Union's ambitious environmental targets [2]. To achieve these goals, various decarbonization alternatives are being proposed in the literature and are already being implemented in ports worldwide to enhance the sustainability

of existing infrastructures and reduce environmental impact [3]–[5].

Port authorities, in their different roles as landlords, regulators, operators, and community managers, can play an important role in implementing and fostering such alternatives; but, as highlighted in [6], this remains a challenge. As pointed out in [5], [7], the applicability of these alternatives to real-world port environments remains limited, since there is a variety of barriers holding ports back from decarbonization implementation. In this context, the first step toward effectively overcoming such barriers is to identify port-specific characteristics and environmental goals, to better guide the selection and prioritization of appropriate alternatives [5]. Ports differ in many aspects, such as geography, operational profile, financing budget, local infrastructure and grid capacity, future development potential, and also regulatory and policy context [5], [6]. Therefore, the unique characteristics of different ports mean that an effective alternative in one context may not be suitable in another, necessitating case-specific analysis to determine the best decarbonization alternative for each port's unique needs.

Despite the existence of diverse studies that assess the effectiveness of various decarbonization solutions, explore their potential and propose implementation schemes in specific contexts (e.g., in short-sea shipping, as detailed in [8]; or the analysis of the potential for transitioning the Port of Souda, Greece, as presented in [9]), port authorities frequently struggle with strategic decision-making. This difficulty stems from the absence of a clear, consistent and unifying framework for the prioritization and sequential implementation of these solutions [9]. The challenge of effective selection is further compounded by the lack of data and decision support tools that would allow port authorities to conduct this essential case-specific analysis in a standardized and time-efficient manner. Even in [9], where the proposed approach provides a structured methodology for integrating decarbonization alternatives into a specific port, the authors recognized the reliance on literature-

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based data and site-relevant benchmarks as limitations, given the absence of direct operational datasets for the chosen case study. To meet this need, as pointed out in [10], several studies have proposed the use of simulation-based methods. However, while operationally accurate, many existing approaches lack detailed energy models or rely on proprietary software and highly detailed data that are often owned by third parties, which significantly limits their usability and adoption for wider stakeholders.

To address these limitations, this work presents an open-source Python-based simulator for accurate energy demand modeling in container terminals. Compared with previous works, the proposed tool accurately models the energy demanded by the different assets at the terminal. This granular approach allows for rapid scenario testing and straightforward comparison of decarbonization alternatives. By coherently modifying the simulator's inputs, users can assess the effect of electrification, implement energy management systems, or introduce efficiency improvement measures. This functionality enables analysis of the impact on the electrical grid and supports financial and economic assessments to prioritize future investments. Furthermore, it allows for the clear evaluation of emissions associated with the activities developed in the port. All evaluated under realistic operational conditions and transparent assumptions, where every calculation and model parameter is visible. This open approach is intended to build trust with engineering teams and stakeholders, facilitating the informed selection and real-world implementation of decarbonization alternatives.

The remainder of this paper is organized as follows: Section II details the architecture and main functionalities of the developed open-source tool; Section III presents an application case study, demonstrating the tool's capability in analyzing potential port decarbonization scenarios. Finally, Section IV concludes the paper.

II. OPSEASIM

The tool presented in this paper (OpSEASim) is a discrete-event simulator (DES) built in Python using the SimPy library. The framework is designed around an agent-based architecture where each physical asset in the terminal is represented as an independent software agent with its own logic and processes that interact with each other. The current version of the simulator (available at <https://oss.inesctec.pt/#/projects?category=Energy>) considers several basic operational processes and physical assets required for container handling, see Fig. 1:

- 1) Vessel arrivals: based on the input data, the time of arrival and vessel's characteristics, as well as the containers to be handled during each call are known;
- 2) OPS connection: if onshore power supply (OPS) is available, the vessel can connect to shore power for the duration of its stay;
- 3) Quay Cranes (QCs) operations: the simulator internally determines the number of QCs committed to each vessel call. This decision is based on a calculation that accounts

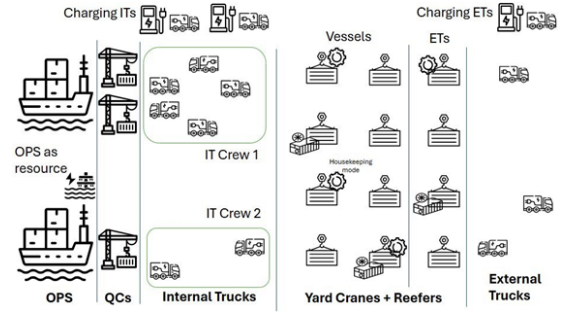


Fig. 1: Container terminal layout considered.

for factors such as cranes availability, the volume of containers to be handled and the specific containership class. The user can readily configure this internal decision-making process to accurately reflect different operational strategies;

- 4) Primary movers: a crew with a fixed number of Internal Trucks (ITs), each with specific performance characteristics, is associated with each QC. This allocation and the vehicle characteristics are easily configurable by the user through the inputs or internally within the simulator. These ITs are responsible for moving containers between the quay-side and the yard storage area;
- 5) Yard operations: the yard configuration assumes that each block is handled by a single Yard Crane (YC). A predefined set of container blocks on the quay-side are specifically associated with each vessel call, interacting directly with the containerships. Conversely, the YCs in the remaining blocks continuously monitor for opportunities to perform housekeeping actions when no External Truck (ET) is appointed to import or export containers from that specific block.
- 6) Battery Electric Vehicle (BEV) chargers: without loss of generality, the model considers the existence of BEV chargers for both ITs and ETs. ETs planning to charge at the terminal premises are serviced on a first-in-first-out basis. Whereas a pre-specified minimum state-of-charge (SOC) is considered for the ITs. Before each new mission, the IT crew agent checks whether the estimated energy demand will cause the SOC to fall below the minimum threshold. If this condition is met, the IT is immediately placed in the queue for charging and replaced, if possible.
- 7) Refrigerated containers (Reefers): the total number of reefers connected within the terminal, along with their specific characteristics, is dynamically determined. This determination is based on the initial yard status and the ongoing import/export processes characterized by vessel calls and ET appointments for the day.

The simulator's architecture, depicted in Fig. 2, is designed with modularity and extensibility in mind, separating the simulation's operational logic from its energy analysis capabilities. The architecture is centered around four main components:

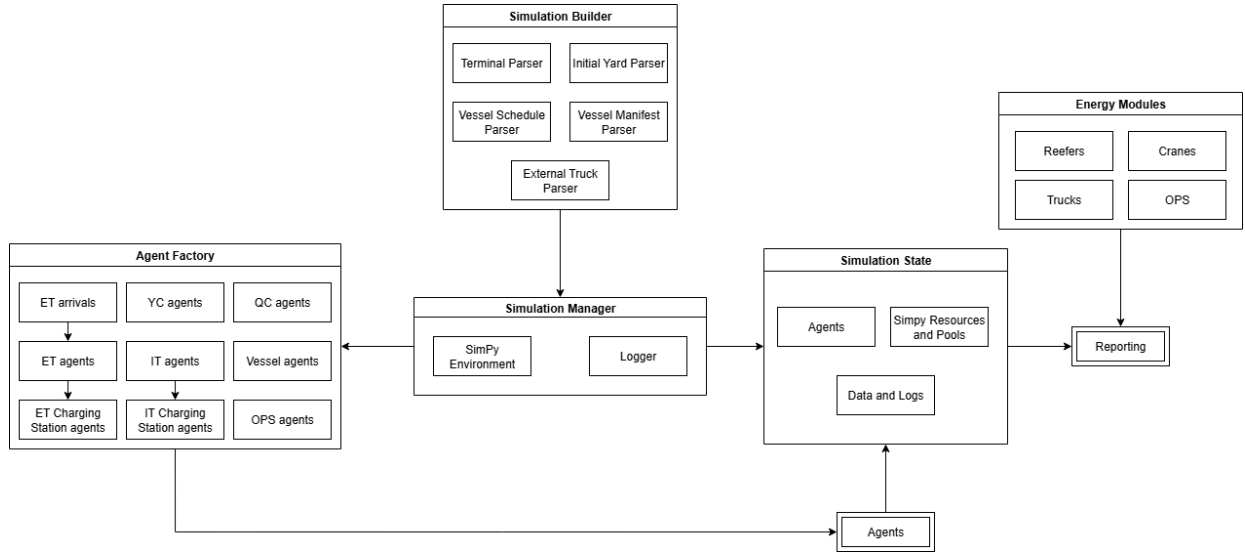


Fig. 2: Architecture of the simulator.

- *Simulation Builder*
- *Simulation Manager*
- *Agent Factory*
- *Simulation State*

The process begins with the *Simulation Builder*, which is responsible for interpreting all user-provided input files. It uses a suite of specialized parsers to configure the simulation environment based on the terminal's physical layout, initial container inventory and operational schedules for vessels and external trucks visiting the terminal. Once the initial configuration is complete, the builder passes control to the *Simulation Manager*. This central component holds the core *SimPy* environment and a dedicated *Logger*. Its primary function is to orchestrate the simulation run. At the start of the simulation, it invokes the *Agent Factory*, which instantiates the entire population of agents (e.g., cranes, vessels, trucks) required for the scenario. These agents interact with each other and with shared resources through a centralized *Simulation State* object. This object acts as a single repository for all shared simulation resources, agent instances and logged data. This design decouples the agents, allowing them to operate autonomously while accessing a consistent, shared world-state.

As the simulation progresses, operational data, such as container movements and equipment usage, is logged in the *Simulation State*. This data serves as the input for the distinct *Energy Modules*. These modules contain the physics-based models used to calculate the energy consumption for different asset classes (reefers, cranes, trucks, vessels and OPS). Finally, a *Reporting* component processes the outputs from both the operational logs and the energy modules to generate the final analysis and visualizations. This modular design ensures that the energy calculations are a distinct layer built upon the operational simulation, allowing for flexibility in analyzing different energy-related scenarios without altering the core logistical framework.

III. CASE STUDY

In this section, we present a use case for the developed tool, focusing specifically on the evaluation of alternative decarbonization scenarios within a container terminal. The analyzed case is inspired by the container terminal at the Port of Sines (Portugal), intending to accurately resemble the terminal's general layout, daily throughput and foreseen decarbonization roadmap.

A. Simulator setup

To run the simulator, several input data files must be provided by the user. These files are used for establishing the terminal layout, the daily schedule and configuring the energy models, as shown in Fig. 3. The input files are provided pre-filled; therefore, if users do not have specific data concerning the assets' energy demand, they can use default values. These default values were obtained from a thorough literature review and can offer an initial baseline for benchmarking.

Fig. 4 illustrates the simulated arrival dynamics for both vessel calls and external trucks scheduled via the terminal's

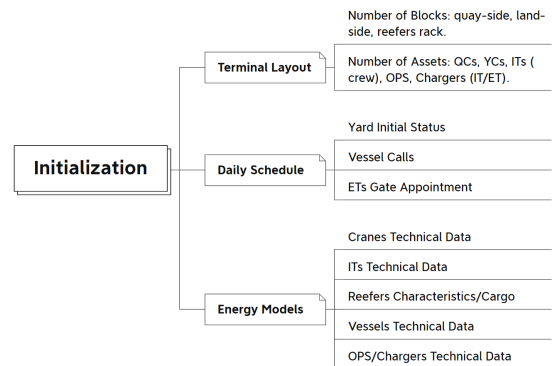


Fig. 3: Simulator initialization.

truck appointment system. As shown in Fig. 4a, the maximum number of containerhips at berth simultaneously during the simulated day (September 26th) is three. Fig. 4b presents a heatmap of the hourly external truck arrivals; two distinct peak periods are observed at approximately 08:00 and 12:00.

B. Scenarios

The performance of the simulator for evaluating decarbonization alternatives is tested by comparing three different scenarios. A complete operational day is simulated based on the provided input data and this Business-as-Usual (BAU) baseline scenario is compared with other two possible scenarios (Mid-El and High-El), representing two different level of electrification at the terminal. As observed in Table I, the main differences between the scenarios are the number of installed OPS units, the level of electrification of the IT fleet, the amount of ETs requiring charging at the terminal parking, and particularly in the highly electrified scenario (High-El), the retrofitting of the QCs to install a regenerative braking systems.

C. Results

The first group of results focuses on operational performance. These metrics can be used to validate the simulator, where any significant discrepancy between simulated and observed real-world values necessitates further model calibration.

For instance, in Table II we present some key performance indicators concerning the utilization of the cranes at the terminal in the three analyzed scenarios. As observed, the utilization rates for both types of cranes align with typical container terminal operations, where yard cranes exhibit higher utilization. Another notable result is the housekeeping ratio. Note that there is no change in operation indicators, considering that the electrification does not really impact operation

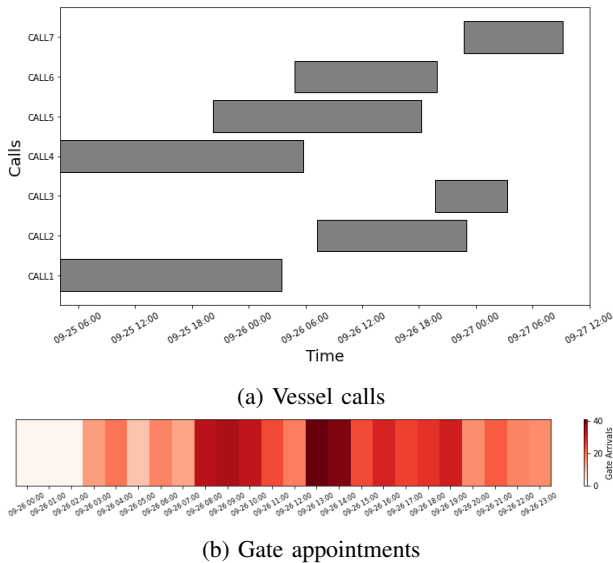


Fig. 4: Arrival patterns at the container terminal.

performance, but rather, just electricity consumption; at least with this terminal layout.

Housekeeping actions involve container movements that cranes must perform to rearrange the yard and ensure that the next container scheduled for export is readily available (i.e., at the top of the stack). Although simulating these movements is complex (requiring the full yard layout, import/export schedules and specific stacking strategies), our simulator maintains a housekeeping-to-total-moves ratio consistent with the 30%–50% range reported in literature [11] and close to the 34% rate observed in real-world operations [12]. Consequently, these results indicate that the simulation results align with real-world container terminal operations. However, as terminal configurations vary significantly, these performance indicators should be validated against specific empirical data from the terminal being modeled.

On the other hand, a key contribution of the proposed tool is its capability to accurately forecast the energy demand of various assets within the terminal. Fig. 5 illustrates the demand associated with shore power connections. In the 1-OPS scenario (Mid-El), a peak power demand of approximately 2 MW is observed, whereas a significantly higher demand of roughly 5 MW occurs when allowing up to three containerhips to draw power from the local electrical grid. One of the main characteristics of the proposed tool is also its ability to represent the considerable volatility of crane demand, see Fig. 6. In contrast to existing approaches, where crane power demand and energy consumption are treated as a fixed values [13], our model accounts for the intrinsic relationship between

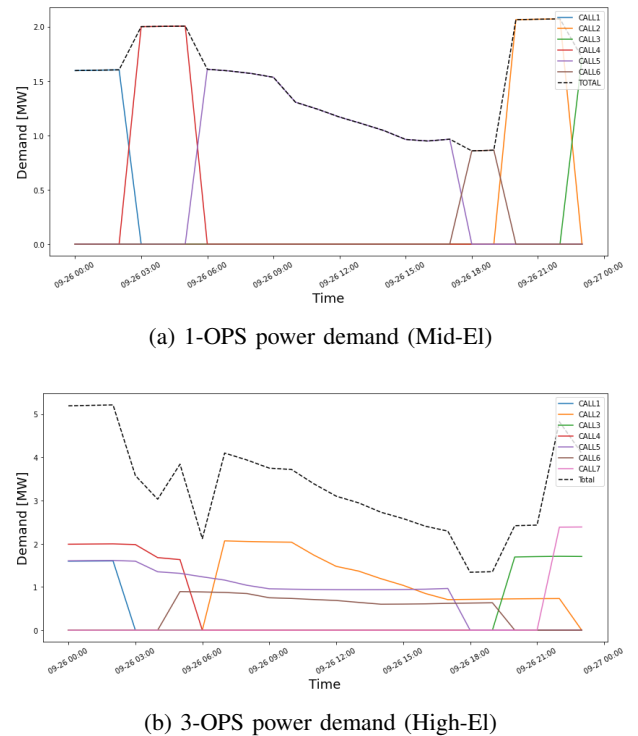


Fig. 5: OPS demand.

TABLE I: Decarbonization scenarios inputs.

Day	Scenarios	Assets						
		QCs	YCs	Reefers	OPS	BEV-IT	BEV-ET	Vessel Calls
Sep 26	BAU	12	17		0	0	0	
	Mid-El	12	17	983	1	50%	5	7
	High-El	12(Reg)	17(Reg)		3	100%	20	

TABLE II: Crane performance indicators.

Type	Indicator	Value
YCs	Total Moves	8454
	Productive Moves	5428
	Re-handles	3026
	Utilization Rate [%]	81.13
	Housekeeping Ratio [%]	35.79
QCs	Total Moves	4983
	Utilization Rate [%]	53.83
	Throughput [moves/min]	3.46

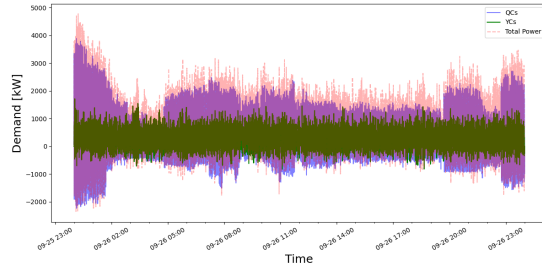


Fig. 6: Cranes power demand.

power demand and container mass, resulting in a more realistic profile. As observed QCs represent the largest consumers; total demand oscillates around 3 MW, with peaks reaching 5 MW during the early hours of the day. Future research will focus on validating the physics-based simulation models through comparison with measured energy consumption data from real terminal operations. Such validation will allow quantification of prediction errors and provide a basis for systematic model calibration.

Regarding the potential decarbonization pathways, several key aspects can be highlighted. First, Fig. 7 illustrates that total final energy demand at the terminal decreases with electrification; this holds true even with the introduction of additional loads, such as external trucks charging on the terminal premises. This reduction comes from the superior efficiency of electric technologies compared to their diesel-driven counterparts. The most prominent example is the substitution of the onboard auxiliary engines with OPS, shifting from maritime diesel oil to electricity, which reduces the energy consumption of berthed containerships by more than two-thirds.

The electrification process also has a positive impact on reducing emissions associated with terminal activities. These benefits are enhanced with a decarbonized grid, as shown in Fig. 8a. Specifically, the highly electrified scenario (High-

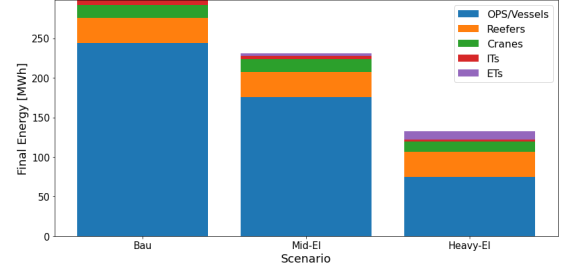
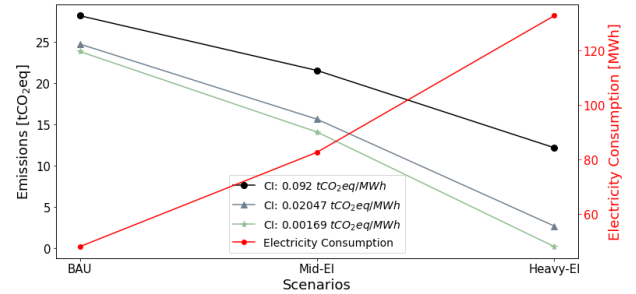
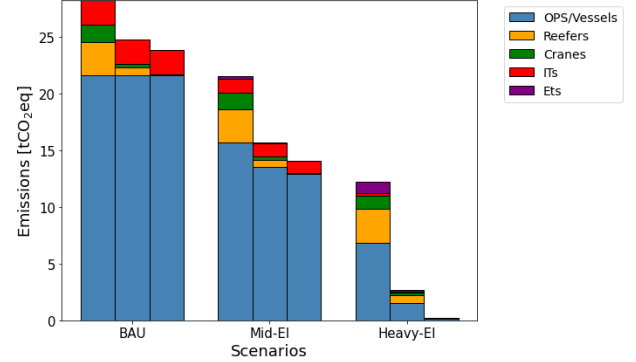


Fig. 7: Final energy by assets and scenarios.



(a) Emissions by scenario and carbon intensity.



(b) Emissions by scenario, carbon intensity and asset type.

Fig. 8: Daily emissions.

El) coupled with a decarbonized electrical grid yields a significant reduction in terminal emissions compared to the BAU scenario. The carbon intensity parameters utilized in this analysis are aligned with the current status and projected trajectories of the Portuguese electrical grid. The baseline grid emission factor is assumed to be 0.092 tCO₂eq/MWh, based on the most recent 2023 data from the Portuguese National Environment Agency [14]. For future projections, values are derived from Portugal's Roadmap for Carbon Neutrality [15],

resulting in emission factors of 20.47×10^{-3} tCO₂eq/MWh and 1.69×10^{-3} tCO₂eq/MWh respectively. Finally, Fig. 8b provides a detailed breakdown of the emissions by asset. As one can observe, the most substantial emission reductions result from the implementation of OPS, with the maximum benefit achieved in the High-El scenario (3 OPS units). The electrification of the IT fleet represents the second most impactful measure. Furthermore, observing the stacked bar charts from left to right reveals the incremental benefits yielded by increasingly decarbonized electrical grids. An important observation is that while this combination of alternatives yields a significant reduction in terminal emissions, such improvements are supported by a substantial increase in the load demand imposed on the existing electrical grid, see Fig. 8a. In fact, the total daily electricity consumption in the terminal increases from approximately 50 MWh in the BAU scenario to 132 MWh in the High-El scenario, representing a more than two-fold rise in energy demand. Therefore, this trade-off underscores the necessity of evaluating grid capacity and potential infrastructure reinforcements when implementing large-scale electrification as decarbonization alternative.

IV. CONCLUSION

In this study, an open-source, multi-agent discrete-event simulator is proposed as an evaluation platform for assessing operational and decarbonization strategies in container terminals. The open-source architecture of the simulator, combined with the granular energy models integrated into the tool, enables rapid and transparent evaluations of alternative strategies, facilitating the analysis of impacts on the electrical grid and supporting financial assessments and the prioritization of future investments.

The performance of the simulator was evaluated by comparing three decarbonization pathways for a medium-sized Portuguese container terminal. The results show that, for this particular port, substantial emissions reductions can be achieved through targeted interventions, notably the deployment of up to three OPS units and the full electrification of the internal truck fleet, combined with a decarbonized electricity generation mix. However, these measures are associated with a significant increase in daily electricity demand, posing challenges for existing grid capacity.

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